

Hypothesis Testing

We want to assess the validity of a claim about the population against a counter-claim using sample data. The two claims are:

Null Hypothesis H_0 is the claim of “no difference.”

Alternative Hypothesis H_1 is the claim we are trying to prove.

Example (Vaccine Trial)

Let P_1 be the fraction of the control group population who get polio.

P_2 is the fraction of the treatment population who get polio.

H_0 is the claim that the vaccine is not effective. $P_1 = P_2$.

H_1 is the claim that the vaccine is effective. $P_1 > P_2$.

The main idea: assume H_0 is true. Does the data contradict the assumption beyond a reasonable doubt?

If yes: accept H_1 and reject H_0 .

If no: H_0 can't be ruled out as an explanation for the data. *In this case we can't accept any hypothesis.*

Analogy: Presumed innocent until proven guilty

H_0 : not guilty

H_1 : guilty

Does the data contradict H_0 ? If yes, then rule as guilty. If no, it doesn't mean the person is innocent but evidence is insufficient to establish guilt, it gives the person the benefit of the doubt.

We can think of a hypothesis test as a method to weigh evidence against H_0 (rather than a decision procedure between H_0 and H_1).

Usually H_0 is chosen to represent a hypothesis of “no difference” between the new and existing methods, just chance alone. If we can reject this sort of hypothesis then we have a **statistically significant** proof of claim H_1 . In this case, the hypothesis test is a **significance test**.

Again, the hypothesis test is not a decision procedure between H_0 and H_1 . If H_1 is not accepted, it does not mean we can accept H_0 . I never want to hear you say: “therefore we accept the null hypothesis”.

A **type I error** is made when a test rejects H_0 in favor of H_1 when H_0 is actually true (false positive).

E.g. person does not have the disease but test is positive.

A **type II error** is made when the test fails to reject H_0 when H_1 is true. (false negative).

E.g. person has the disease but test is negative.

	Decision	
	Do not reject H_0	Reject H_0
H_0 is true	Correct Decision	Type I Error
H_0 is false	Type II Error	Correct Decision

The probability of a type I error is called the **α -risk**.

The probability of a type II error is called the **β -risk**.

That is:

$$\alpha = P(\text{type I error}) = P(\text{reject } H_0 \mid H_0)$$

$$\beta = P(\text{type II error}) = P(\text{fail to reject } H_0 \mid H_1).$$

Type I error is sometimes more serious than Type II, e.g., Type I is like convicting an innocent person and Type II is letting a guilty person go free due to lack of evidence.

Here we create a test that is required to satisfy $P(\text{type I error}) \leq \alpha$. In this case, α is called the **level of significance**, and the test is a **α -level test**.

Usually we choose $\alpha = 0.05$ (or 0.10 or 0.01). This says that most of the time, we accept less than 5% probability of Type I error.

E.g., if we are checking whether: $P(\text{type I error}) = P(\text{test rejects } H_0 \mid H_0) \leq 0.10$, then the test is a 0.10-level test.

The **p-value** is the probability of observing a sample statistic *as extreme or more extreme* than the one observed under the assumption that the null hypothesis is true. It is the “observed level of significance.”

When testing a hypothesis, state α . Calculate the p-value and if the p-value $\leq \alpha$, then reject H_0 . Otherwise do not reject H_0 .

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15.075J / ESD.07J Statistical Thinking and Data Analysis
Fall 2011

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